



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester I 2025/2026

Subject : Enterprise Systems Design & Modelling (SECP3744)
Task : **Technical Report**
Due : 24/10/2025 (Friday)
Submission : Upload softcopy short paper (in .pdf format) via e-learning.
Assignment Type : Group assignment (in pairs)

Objectives:

This assignment aims to help you to develop professional writing, analytical thinking, and responsible use of AI tools in producing a structured, evidence-based report.

Your technical paper should demonstrate your ability to:

- Analyze and communicate ideas about enterprise systems effectively.
- Write professionally using structured, evidence-based arguments.
- Use AI tools ethically and transparently in the research process.

Choose 1 topic and prepare a Technical Report.

Topics:

1. The Role of Artificial Intelligence in Enhancing ERP Decision-Making
2. How AI-powered analytics improve forecasting, resource allocation, and customer service.
3. Blockchain for Supply Chain Transparency and Trust in Enterprise Systems
4. Using blockchain for traceability, fraud prevention, and smart contracts.
5. Cloud-Based ERP vs On-Premise ERP: Future Directions for Enterprises
6. Predictive Analytics in Enterprise Information Systems for Strategic Advantage
7. Using AI in ERP for Demand Forecasting
8. Robotic Process Automation (RPA) in Enterprise Systems
9. Automating repetitive business processes to improve efficiency and reduce errors.
10. Cybersecurity Challenges and Ethical Concerns in Next-Generation Enterprise Systems
11. Risks from AI, IoT integration, and data governance in global enterprises.
12. The Convergence of IoT and ERP: Creating the Intelligent Enterprise
13. Business Intelligence (BI) Dashboards and Visual Analytics in ERP Systems
14. Using AI, blockchain, and analytics to promote sustainability reporting and green supply chains.
15. The Future of Work with Enterprise IS: Low-Code/No-Code Platforms and Democratization of Development

Instructions

1. Format and Structure

The paper must follow the **Technical Report Template** provided. The report should include the following sections:

- Executive Summary
- Introduction
- Background / Literature Review
- Analysis and Discussion
- Recommendations
- Conclusion
- References
- Appendix / Logbook

2. Length

The paper should be 3–5 pages, excluding the title page and references.

3. Content Requirements

Use your creativity to organize the content according to your chosen topic. Include comparative analysis, examples, and discussion of advantages and challenges where relevant. The report should be written in a professional, clear, and concise tone suitable for a business or technical audience.

4. References and Citation

Cite at least seven (7) credible references (journal articles, reports, or trusted online sources). Use APA citation style. Include all sources in the References section at the end of the paper.

Give credit to any figures, tables, or images used from other resources.

5. Use of AI Tools

You are encouraged to use AI tools (e.g., ChatGPT, Gemini, Copilot) to support your research process. However, AI tools must be used responsibly — for idea generation, structure planning, or summarization only. All AI usage must be documented in the Appendix/Logbook, including:

- Date and time used
- Prompt(s) given
- Verification steps to ensure accuracy.

6. Logbook / Process Record

Keep a simple log of your research process: keywords used, AI prompts, key references found, and progress updates. You may record this in GitHub, Google Docs, or a handwritten logbook, and include a summary or screenshot in the Appendix.

7. Plagiarism and Academic Integrity

Plagiarism is strictly prohibited. If any content is directly copied from websites, blogs, or articles without proper citation, a ZERO (0) mark will be given. Include your Turnitin report upon submission. You may use information or visuals from other resources, but always provide proper credits and citations.

8. Submission

Submit a PDF file through e-Learning before the due date. File name format:
GroupNumber _TechnicalReport.pdf

9. Assessment Weightage

This assignment contributes **10% of the total course marks**. Marks will be awarded based on the Technical Writing Rubric provided.

Technical Writing Assignment Template

Executive Summary

Provide a concise summary (approximately 150–200 words) highlighting the paper's purpose, key findings or arguments, and main conclusions or recommendations. Write this section last.

1. Introduction

Introduce the topic and explain its relevance to enterprise systems. Provide background or context and state the objectives, scope, and focus of the paper. (≈300 words)

2. Background / Literature Review

Summarize prior studies, emerging trends, or relevant technologies related to the topic. Include at least 3–5 credible references. Demonstrate understanding of current developments or research gaps. (≈300–400 words)

3. Analysis and Discussion

Discuss the main theme or technology in detail, including:

- How the technology works or functions
- Case examples or applications
- Advantages and disadvantages
- Challenges and future trends

Use visuals (figures, tables, diagrams) where appropriate. (≈500–600 words)

4. Recommendations

Provide practical or strategic suggestions based on your analysis. Explain how organizations or practitioners can apply your findings. (≈200–300 words)

5. Conclusion

Summarize key takeaways and insights. End with a forward-looking statement, highlighting lessons learned or possible future research directions. (≈150–200 words)

References

List at least seven (7) credible references using a the APA format.

Appendix / Logbook

Attach your record of the research process, including AI prompts used (if any), verification steps, keywords, and tools consulted. Screenshots from GitHub or logbook entries can be included here.

Rubric for Technical Writing Assignment (100 Marks)

Criteria	Excellent (10–8)	Good (7–5)	Poor (4–0)	Weight
Structure & Organization	Complete, logical flow; all sections integrated	Mostly organized; minor issues	Disorganized or missing sections	10
Executive Summary	Clear, concise, reflects full paper	Covers main points but lacks clarity	Unclear or missing	5
Introduction	Strong background and clear purpose	Some context but lacks clarity	Weak or missing focus	10
Background / Literature Review	Synthesizes relevant and recent sources	Includes sources but limited analysis	Minimal or irrelevant	10
Analysis & Discussion	Deep and balanced; uses examples and visuals	Adequate but surface-level	Descriptive or shallow	15
Recommendations	Insightful, practical, and supported	Adequate but generic	Weak or absent	10
Conclusion	Strong, clear, concise	Acceptable summary	Vague or disconnected	5
Writing Quality & Mechanics	Professional tone; error-free	Minor errors; generally clear	Poor grammar; unclear	10
References & Citation	≥7 credible sources; accurate formatting	Some minor errors or missing elements	Few or incorrect sources	10
Logbook / Process Record	Comprehensive record of process and tools	Acceptable record	Missing or incomplete	5
Timeliness	Submitted on or before due date	1-day late	>1 day late	5
Responsible Use of AI Tools	Documented prompts; verified outputs; ethical use	Some documentation or unclear verification	No record or unethical use	5